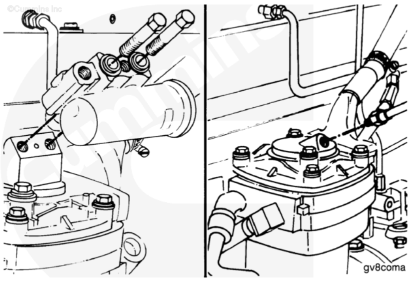
Trainings

Initial Check

⚠ WARNING ⚠

Air pressure must be released from the system before removing the air governor. The governor can be under pressure and cause personal injury.

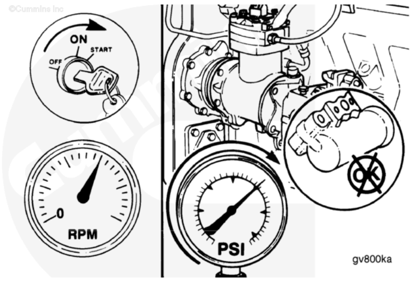


Note : The illustrations shown will be of the SS model single cylinder air compressor. Differences in procedures for SS, QE, and ST model Holset® air compressors will be shown, where necessary.

Remove the air governor or air governor hose from the air compressor unloader body.

Operate the engine to activate the air compressor.

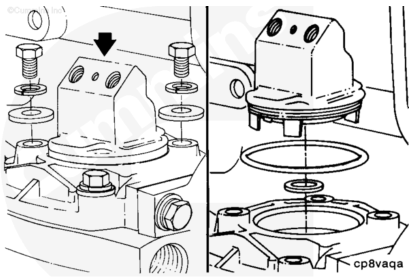
If the air compressor is **not** pumping, the unloader valve is malfunctioning, and **must** be repaired or replaced.



Remove

⚠ WARNING ⚠

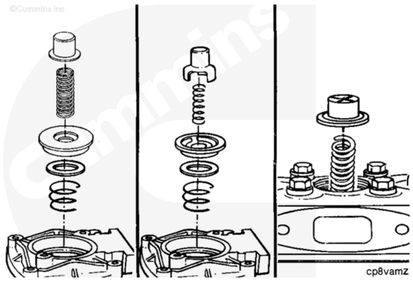
Wear appropriate eye and face protection when performing this procedure. Springs are under tension and can act as a projectile if released, causing personal injury.



Holset® SS, E-Type, and ST Models

- Hold the unloader valve body down and remove the two captive washer capscrews and the two plain washers.
- Remove the unloader valve body.
- Remove the o-ring seal.
- Remove the rectangular ring seal.

Remove the unloader valve cap and the unloader valve spring.

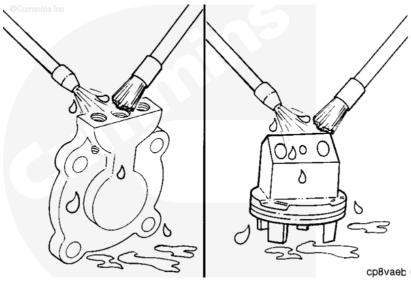


Note : Disassembly of the center unloader valve on Holset® two-cylinder air compressors is similar to the single-cylinder unloader valve.

Remove the intake valve seat and valve.

Remove the intake valve spring.

Clean

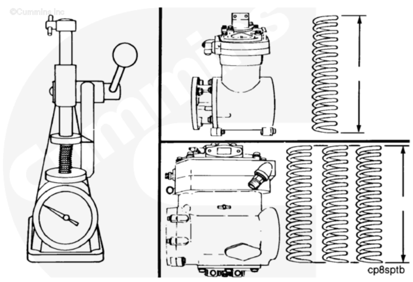


⚠ CAUTION ⚠

Do not use caustic cleaners. Damage to the component can occur.

Remove all carbon and varnish from the unloader valve cap body.

Inspect for Reuse



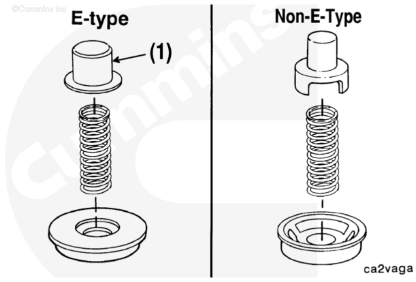
Use a valve spring tester to check the unloader springs.

Use the Holset® Air Compressors Master Repair Manual, Bulletin 3666121, for the spring specifications.

Replace the unloader spring if it does **not** meet the specifications shown, or the wrong spring has been used.

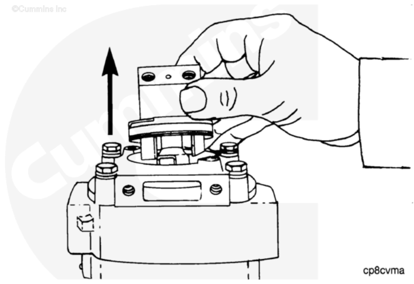
Note : For Holset® two-cylinder air compressors, check both cylinder and center unloader springs. Holset® Engineering Co., Inc. recommends that new springs be installed during rebuild.

Note : If the compressor has a flat hat-type unloader cap (1), it **must** use an unloader spring and valve seat different from that used with the three-prong unloader.



⚠ WARNING ⚠

Wear appropriate eye and face protection when performing this procedure. Springs are under tension and can act as a projectile if released, causing personal injury.

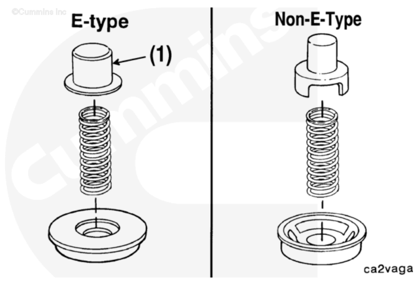


Air Compressor Intake - Inspect

- Remove the capscrews, the lock washers, and the flat washers that secure the unloader valve assembly to the cylinder head cover.
- Remove the unloader valve assembly and the spring from the cylinder head and the cover.

The air compressor is built with one of the two types of unloader valves. One is referred to as a flat hat type and the other is a three-prong.

The cleaning procedure is the same for both.

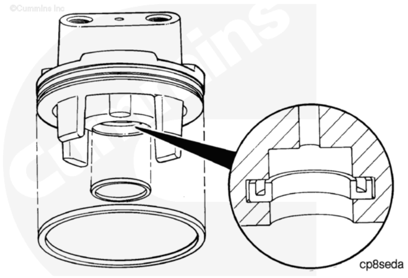


Inspect the unloader valve for carbon buildup.

If carbon or heavy varnish is present, remove, clean, and inspect the compressor head and the valve assembly. Replace parts as necessary. Use the Holset® Air Compressors Master Repair Manual, Bulletin 3666121, for procedures or contact a Cummins® Authorized Repair Location.

If the unloader valve is clean or **only** lightly varnished, install a new o-ring on the unloader body and a new rectangular seal inside the unloader body cavity.

The open side of the rectangular seal **must** face the top of the unloader body.

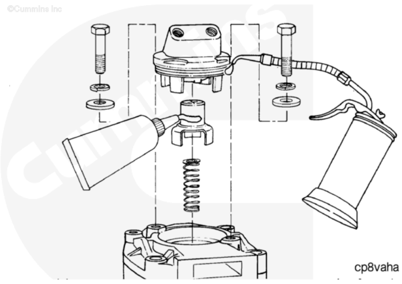


Lubricate the unloader cap with anti-seize compound.
Lubricate the unloader body o-ring with engine oil.

Assemble the unloader assembly to the cylinder head cover.

Tighten the capscrews.

Torque Value: 14 n•m [124 in-lb]



Install

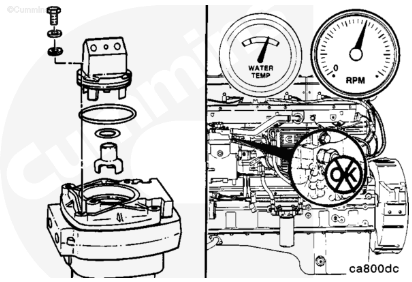
Holset® SS, E-Type, and ST Models

Assemble the air compressor.

Grease the rectangular ring seal, unloader cap, and unloader body bore with high-temperature grease (Accrolube lubrication Teflon® grease or equivalent).

Torque Value: 14 n•m [124 in-lb]

Start and operate the engine and check the compressor for air leaks and proper operation.



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